

**9** A transformation  $T$  of the plane is represented by the matrix  $\mathbf{M} = \begin{pmatrix} k+1 & -1 \\ 1 & k \end{pmatrix}$ , where  $k$  is a constant.

Show that, for all values of  $k$ ,  $T$  has no invariant lines through the origin. **[6]**